Homework 12.10.26 – Exercises 9–18

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Exercise 9

Given $\mu I \preceq \nabla^2 f \preceq LI$, $0 < \mu \leq L$, and $T_\gamma(x) := x - \gamma \nabla f(x)$.

1)

$$T_\gamma(x) - T_\gamma(y) = \int_0^1 \nabla T_\gamma\big(y + t(x-y)\big)(x-y)\, dt \;\Rightarrow$$

$$\|T_\gamma(x) - T_\gamma(y)\| = \left\| \int_0^1 \nabla T_\gamma\big(y + t(x-y)\big)(x-y)\, dt \right\| \leq \int_0^1 \left\|\nabla T_\gamma\big(y + t(x-y)\big)\right\| \|x-y\|\, dt$$
$$\leq \int_0^1 \sup_{z \in \mathbb{R}^d} \|\nabla T_\gamma(z)\|\, dt \cdot \|x-y\| = \sup_{z \in \mathbb{R}^d} \left\|I - \gamma \nabla^2 f(z)\right\| \|x-y\|.$$

Since $\mu I \preceq \nabla^2 f \preceq LI$, the eigenvalues λ_i of the symmetric matrix $\nabla^2 f(z)$ lie in $[\mu, L]$, so

$$\left\|I - \gamma \nabla^2 f(z)\right\| = \max_i |1 - \gamma \lambda_i| \leq \max\{|1-\gamma L|, |1-\mu\gamma|\}$$

($|1-\gamma\lambda|$ is convex in λ, so its maximum on $[\mu, L]$ is at an endpoint) $\Rightarrow$

$$\|T_\gamma(x) - T_\gamma(y)\| \leq \|x-y\| \cdot \max\{|1-\gamma L|, |1-\mu\gamma|\}.$$

2)

If $\gamma = \frac{2}{\mu+L} \Rightarrow |1-\gamma L| = |1-\mu\gamma| = \frac{L-\mu}{L+\mu}$.

If $\gamma > \frac{2}{\mu+L} \Rightarrow L\gamma > \frac{2L}{\mu+L} \Rightarrow 1 - L\gamma < 1 - \frac{2L}{\mu+L} = \frac{\mu-L}{\mu+L} \leq 0 \Rightarrow |1-L\gamma| > \frac{L-\mu}{L+\mu} \Rightarrow \max\{|1-\gamma L|, |1-\mu\gamma|\} > \frac{L-\mu}{L+\mu}$.

If $\gamma < \frac{2}{\mu+L} \Rightarrow \mu\gamma < \frac{2\mu}{\mu+L} \Rightarrow 1 - \mu\gamma > 1 - \frac{2\mu}{\mu+L} = \frac{L-\mu}{L+\mu} \geq 0 \Rightarrow |1-\mu\gamma| > \frac{L-\mu}{L+\mu} \Rightarrow \max\{|1-\gamma L|, |1-\mu\gamma|\} > \frac{L-\mu}{L+\mu}$.

$\Rightarrow \gamma = \frac{2}{\mu+L}$ is optimal.

3)

Since $\nabla f(x^\star) = 0$, we have $T_\gamma(x^\star) = x^\star$, so with $\gamma = \frac{2}{\mu+L}$

$$\left\|x^{k+1} - x^\star\right\| = \left\|T_\gamma(x^k) - T_\gamma(x^\star)\right\| \leq \frac{L-\mu}{L+\mu} \left\|x^k - x^\star\right\| \;\Rightarrow\; \left\|x^k - x^\star\right\| \leq \|x^0 - x^\star\| \cdot \left(\frac{L-\mu}{L+\mu}\right)^k.$$

In Theorem 15, $\left\|x^k - x^\star\right\|^2 \leq (1-\gamma\mu)^k \left\|x^0 - x^\star\right\|^2$, where $\gamma \leq \frac{1}{L}$; the best choice is $\gamma = \frac{1}{L}$

$$\Rightarrow \left\|x^k - x^\star\right\|^2 \leq \left(1 - \frac{\mu}{L}\right)^k \|x^0 - x^\star\|^2 = \left(\frac{L-\mu}{L}\right)^k \|x^0 - x^\star\|^2.$$

Squaring our bound, $\left\|x^k - x^\star\right\|^2 \leq \left(\frac{L-\mu}{L+\mu}\right)^{2k} \left\|x^0 - x^\star\right\|^2$, and

$$\frac{L-\mu}{L} : \left(\frac{L-\mu}{L+\mu}\right)^2 = \frac{(L+\mu)^2}{L(L-\mu)} > 1 \quad (\mu < L),$$

so the rate with $\gamma = \frac{2}{\mu+L}$ is better, by the factor $\frac{(L+\mu)^2}{L(L-\mu)}$ per iteration.

Exercise 10

If f is L-smooth and μ-strongly convex, then by (18) and Exercise 8

$$\frac{1}{L}\left\|\nabla f(x^k) - \nabla f(x^\star)\right\|^2 \leq 2D_f(x^k, x^\star) \leq \frac{1}{\mu}\left\|\nabla f(x^k) - \nabla f(x^\star)\right\|^2.$$

Dividing by $\left\|\nabla f(x^k) - \nabla f(x^\star)\right\|^2$, (i) and (ii) are true: $\frac{1}{L} \leq \gamma_k \leq \frac{1}{\mu}$.

For (iii): by (i), $1 - \gamma_k L \leq 0$, so the last term of Lemma 14 cannot be dropped. Instead we stop the proof of Lemma 14 before using (25), with $r^k = x^k - x^\star$:

$$\left\|r^{k+1}\right\|^2 \leq (1-\gamma_k\mu)\left\|r^k\right\|^2 - 2\gamma_k D_f(x^k, x^\star) + \gamma_k^2\left\|\nabla f(x^k) - \nabla f(x^\star)\right\|^2.$$

By the definition of γ_k, $\gamma_k^2\left\|\nabla f(x^k) - \nabla f(x^\star)\right\|^2 = 2\gamma_k D_f(x^k, x^\star) \Rightarrow$

$$\left\|x^{k+1} - x^\star\right\|^2 \leq (1-\gamma_k\mu)\left\|x^k - x^\star\right\|^2.$$

Exercise 11

$f : \mathbb{R}^d \to \mathbb{R}$ is differentiable. Then for all $x, y, z \in \mathbb{R}^d$

$$D_f(x,z) = D_f(x,y) + \langle x - y, \nabla f(y) - \nabla f(z)\rangle + D_f(y,z). \tag{a}$$

This is the same as

$$\begin{aligned} f(x) - f(z) - \langle \nabla f(z), x - z\rangle = {} & f(x) - f(y) - \langle \nabla f(y), x - y\rangle \\ & + \langle x - y, \nabla f(y) - \nabla f(z)\rangle + f(y) - f(z) - \langle \nabla f(z), y - z\rangle. \end{aligned}$$

Cancel the common terms on both sides ($f(\cdot)$):

$$-\langle \nabla f(z), x - z\rangle = -\langle \nabla f(y), x - y\rangle - \langle \nabla f(z), y - z\rangle + \langle x - y, \nabla f(y) - \nabla f(z)\rangle.$$

Also $\langle x - y, \nabla f(y) - \nabla f(z)\rangle = \langle \nabla f(y), x - y\rangle - \langle \nabla f(z), x - y\rangle$, so this is the same as

$$-\langle \nabla f(z), x - z\rangle = -\langle \nabla f(z), y - z\rangle - \langle \nabla f(z), x - y\rangle$$

$\iff$

$$\langle -\nabla f(z), x - z + z - y + y - x\rangle = \langle -\nabla f(z), 0\rangle = 0,$$

which is true.

(i)

If $x = z$, then $D_f(x,z) = f(x) - f(x) - \langle \nabla f(x), x - x\rangle = 0$. As a result we get

$$D_f(x,y) + D_f(y,x) = -\langle x - y, \nabla f(y) - \nabla f(x)\rangle = \langle \nabla f(x) - \nabla f(y), x - y\rangle.$$

(ii)

If $f(x) = \frac{1}{2}\|x\|^2$, then $\nabla f(v) = v \Rightarrow \langle x - y, \nabla f(y) - \nabla f(z)\rangle = \langle x - y, y - z\rangle$. Also

$$\begin{aligned} 2D_f(x,y) &= \|x\|^2 - \|y\|^2 - 2\langle y, x - y\rangle = \langle x, x\rangle - \langle y, x\rangle - \langle y, x - y\rangle \\ &= \langle x - y, x\rangle - \langle y, x - y\rangle = \langle x - y, x - y\rangle = \|x - y\|^2 \;\Rightarrow\; D_f(x,y) = \tfrac{1}{2}\|x - y\|^2 . \end{aligned}$$

As a result, for $f(x) = \frac{1}{2}\|x\|^2$, (a) becomes

$$\tfrac{1}{2}\|x - z\|^2 = \tfrac{1}{2}\|x - y\|^2 + \langle x - y, y - z\rangle + \tfrac{1}{2}\|y - z\|^2 .$$

Exercise 12

$f : \mathbb{R}^d \to \mathbb{R}$, $f \in C^2(\mathbb{R}^d)$. Then f is convex if and only if $\nabla^2 f(x) \succeq 0$.

1)

Assume $\nabla^2 f(x) \succeq 0$ for all x. Then

$$\langle \nabla f(x) - \nabla f(y), x - y\rangle = \int_0^1 \big\langle \nabla^2 f\big(y + t(x - y)\big)(x - y), x - y\big\rangle \, dt.$$

With $v := x - y$ we get $\int_0^1 \langle \nabla^2 f(y + tv)\, v, v\rangle \, dt$, and because $\nabla^2 f \succeq 0$, $\langle \nabla^2 f(y + tv)\, v, v\rangle \geq 0$. As a result $\langle \nabla f(x) - \nabla f(y), x - y\rangle \geq 0$, so f is convex by Proposition 7 ((iii) $\Rightarrow$ (i)).

2)

If f is convex, then $f(x) \geq f(y) + \langle \nabla f(y), x - y\rangle$. Define $x := y + hv$, $h \in \mathbb{R}$, $v \in \mathbb{R}^d$. Then

$$f(y + hv) \geq f(y) + h\langle \nabla f(y), v\rangle .$$

Also, because $f \in C^2(\mathbb{R}^d)$,

$$f(y + hv) = f(y) + h\langle \nabla f(y), v\rangle + \frac{h^2}{2}\langle \nabla^2 f(y)\, v, v\rangle + o(h^2\|v\|^2).$$

As a result $\frac{h^2}{2}\langle \nabla^2 f(y)\, v, v\rangle + o(h^2\|v\|^2) \geq 0$ for all h, v, y. Dividing by $\frac{h^2}{2}$ and letting $h \to 0$: $\langle \nabla^2 f(y)\, v, v\rangle \geq 0 \Rightarrow \nabla^2 f(y) \succeq 0$ for all y.

Exercise 13

$f : \mathbb{R}^d \to \mathbb{R}$ is convex, $x_i \in \mathbb{R}^d$, $\lambda_i \geq 0$ and $\sum_{i=1}^m \lambda_i = 1$. Then $f\big(\sum_{i=1}^m \lambda_i x_i\big) \leq \sum_{i=1}^m \lambda_i f(x_i)$.

1)

For $m = 2$, $\lambda_1 := \lambda$, then $\lambda_2 = 1 - \lambda$, and $f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda) f(x_2)$ by convexity.

2)

If the inequality is true for $m = k$, we show it for $m = k+1$. If $\lambda_k + \lambda_{k+1} = 0$, then $\lambda_k = \lambda_{k+1} = 0$ and this is the case $m = k$. Otherwise let

$$\lambda'_k := \lambda_k + \lambda_{k+1}, \qquad x'_k := \frac{\lambda_k x_k + \lambda_{k+1} x_{k+1}}{\lambda_k + \lambda_{k+1}},$$

and $\lambda'_i := \lambda_i$, $x'_i := x_i$ for $i = 1, \dots, k-1$. Then

$$f\Big(\sum_{i=1}^{k+1} \lambda_i x_i\Big) = f\Big(\sum_{i=1}^{k} \lambda'_i x'_i\Big) \le \sum_{i=1}^{k} \lambda'_i f(x'_i) = \sum_{i=1}^{k-1} \lambda_i f(x_i) + (\lambda_k + \lambda_{k+1})\, f\Big(\frac{\lambda_k x_k + \lambda_{k+1} x_{k+1}}{\lambda_k + \lambda_{k+1}}\Big). \quad (1)$$

By the case $m = 2$,

$$f\Big(\frac{\lambda_k}{\lambda_k + \lambda_{k+1}} x_k + \frac{\lambda_{k+1}}{\lambda_k + \lambda_{k+1}} x_{k+1}\Big) \le \frac{\lambda_k}{\lambda_k + \lambda_{k+1}} f(x_k) + \frac{\lambda_{k+1}}{\lambda_k + \lambda_{k+1}} f(x_{k+1}). \quad (2)$$

From (1) and (2),

$$f\Big(\sum_{i=1}^{k+1} \lambda_i x_i\Big) \le \sum_{i=1}^{k-1} \lambda_i f(x_i) + \lambda_k f(x_k) + \lambda_{k+1} f(x_{k+1}) = \sum_{i=1}^{k+1} \lambda_i f(x_i).$$

Exercise 14

Let $f : \mathbb{R}^d \to \mathbb{R}$, $f \in C^1(\mathbb{R}^d)$. Then the following statements are equivalent:

(i) f is μ-convex

(ii) $\mu \|x - y\|^2 \le 2D_f(x, y)$ for all $x, y \in \mathbb{R}^d$

(iii) $\mu \|x - y\|^2 \le \langle \nabla f(x) - \nabla f(y), x - y \rangle$ for all $x, y \in \mathbb{R}^d$

(i) ⇒ (ii)

$g(x) := f(x) - \frac{\mu}{2} \|x\|^2$, then g is convex, as a result (Proposition 7)

$$D_g(x, y) = g(x) - g(y) - \langle \nabla g(y), x - y \rangle \ge 0,$$

which is the same as

$$f(x) - f(y) - \langle \nabla f(y), x - y \rangle - \mu\Big(\frac{\|x\|^2}{2} - \frac{\|y\|^2}{2} - \langle y, x - y \rangle\Big) \ge 0.$$

On the other hand $\frac{\|x\|^2}{2} - \frac{\|y\|^2}{2} - \langle y, x - y \rangle = \frac{1}{2} \|x - y\|^2$. And finally we get

$$f(x) - f(y) - \langle \nabla f(y), x - y \rangle \ge \frac{\mu}{2} \|x - y\|^2 .$$

(ii) ⇒ (iii)

$$\begin{aligned} & f(x) - f(y) - \langle \nabla f(y), x - y \rangle \ge \tfrac{\mu}{2} \|x - y\|^2 \\ + \; & f(y) - f(x) - \langle \nabla f(x), y - x \rangle \ge \tfrac{\mu}{2} \|x - y\|^2 \\ = \; & - \langle \nabla f(y), x - y \rangle + \langle \nabla f(x), x - y \rangle \ge \mu \|x - y\|^2 , \end{aligned}$$

which is the same as $\langle \nabla f(x) - \nabla f(y), x - y \rangle \ge \mu \|x - y\|^2$.

(iii) ⇒ (i)

First, if $\langle \nabla g(x) - \nabla g(y), x - y \rangle \geq 0$ for all x, y, then g is convex:

$$g(y) = g(x) + \int_0^1 \big\langle \nabla g\big(x + t(y-x)\big), y - x \big\rangle \, dt.$$

With $u := x + t(y-x)$, $t \in (0,1]$, we get $\langle \nabla g(u) - \nabla g(x), u - x \rangle \geq 0$. Expanding, $\langle \nabla g(u), u - x \rangle \geq \langle \nabla g(x), u - x \rangle$, and since $u - x = t(y-x)$, dividing by t:

$$\langle \nabla g(u), y - x \rangle \geq \langle \nabla g(x), y - x \rangle .$$

Integrating over t,

$$\int_0^1 \big\langle \nabla g\big(x + t(y-x)\big), y - x \big\rangle \, dt \geq \int_0^1 \langle \nabla g(x), y - x \rangle \, dt = \langle \nabla g(x), y - x \rangle ,$$

as a result $g(y) - g(x) \geq \langle \nabla g(x), y - x \rangle$, so g is convex by Proposition 7 ((ii) ⇒ (i)).

Now let $g(x) := f(x) - \frac{\mu}{2} \|x\|^2$, so $\nabla g(x) = \nabla f(x) - \mu x$. Then

$$\langle \nabla g(x) - \nabla g(y), x - y \rangle = \langle \nabla f(x) - \nabla f(y) - \mu(x - y), x - y \rangle \geq 0$$

is the same as $\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq \mu \|x - y\|^2$. So (iii) gives that g is convex, which is the same as f being μ-convex.

Exercise 15

If $f : \mathbb{R}^d \to \mathbb{R}$, $f \in C^1(\mathbb{R}^d)$ and f is μ-convex, then

(ii) $2D_f(x, y) \leq \frac{1}{\mu} \|\nabla f(x) - \nabla f(y)\|^2$ for all $x, y \in \mathbb{R}^d$

(iii) $\langle \nabla f(x) - \nabla f(y), x - y \rangle \leq \frac{1}{\mu} \|\nabla f(x) - \nabla f(y)\|^2$ for all $x, y \in \mathbb{R}^d$

(iii)

Let's assume for a moment that (ii) is true (proved below). Then we have

$$\langle \nabla f(x) - \nabla f(y), x - y \rangle = f(x) - f(x) + f(y) - f(y) + \langle \nabla f(x), y - x \rangle + \langle \nabla f(y), x - y \rangle .$$

Regrouping them,

$$\begin{aligned}
&= \big(f(x) - f(y) - \langle \nabla f(y), x - y \rangle\big) + \big(f(y) - f(x) - \langle \nabla f(x), y - x \rangle\big) \\
&= D_f(x, y) + D_f(y, x) \leq \frac{1}{2\mu} \|\nabla f(x) - \nabla f(y)\|^2 + \frac{1}{2\mu} \|\nabla f(y) - \nabla f(x)\|^2 \\
&= \frac{1}{\mu} \|\nabla f(x) - \nabla f(y)\|^2 .
\end{aligned}$$

(ii)

First of all let's define $a := \nabla f(x) - \nabla f(y)$, $b := x - y$. Then

$$0 \leq \left\| \frac{a}{\sqrt{\mu}} - \sqrt{\mu}\, b \right\|^2 = \frac{1}{\mu} \|a\|^2 - 2 \langle a, b \rangle + \mu \|x - y\|^2 ,$$

which means

$$\frac{1}{\mu} \|\nabla f(x) - \nabla f(y)\|^2 \geq 2 \langle \nabla f(x) - \nabla f(y), x - y \rangle - \mu \|x - y\|^2 . \tag{1}$$

On the other hand (Proposition 10, (ii))

$$f(x) - f(y) - \langle \nabla f(y), x - y \rangle \geq \frac{\mu}{2} \|x - y\|^2,$$

which is the same as

$$f(y) - f(x) \leq \langle \nabla f(y), y - x \rangle - \frac{\mu}{2} \|x - y\|^2,$$

also

$$f(y) - f(x) - \langle \nabla f(x), y - x \rangle \leq \langle \nabla f(y) - \nabla f(x), y - x \rangle - \frac{\mu}{2} \|x - y\|^2 \overset{(1)}{\leq} \frac{1}{2\mu} \|\nabla f(x) - \nabla f(y)\|^2,$$

so $D_f(y, x) \leq \frac{1}{2\mu} \|\nabla f(x) - \nabla f(y)\|^2$. As a result $2D_f(y, x) \leq \frac{1}{\mu} \|\nabla f(x) - \nabla f(y)\|^2$ for all $x, y \in \mathbb{R}^d$, and swapping the names of x and y gives (ii).

Exercise 16

If $f : \mathbb{R}^d \to \mathbb{R}$, $f \in C^2(\mathbb{R}^d)$, then f is μ-convex iff $\mu I \preceq \nabla^2 f(x)$ for all x.
$g(x) := f(x) - \frac{\mu}{2} \|x\|^2$, then $g \in C^2(\mathbb{R}^d)$ and $\nabla^2 g(x) = \nabla^2 f(x) - \mu I$.

1)

If $\nabla^2 f(x) \succeq \mu I$, then $\nabla^2 g(x) \succeq 0$, then g is convex (Exercise 12), then f is μ-convex.

2)

If f is μ-convex, then g is convex, then $\nabla^2 g(x) \succeq 0$ (Exercise 12), then $\nabla^2 f(x) \succeq \mu I$.

Exercise 17

Let $f : \mathbb{R}^d \to \mathbb{R}$ be continuously differentiable. Then the following statements are equivalent:

(i) f is L-smooth

(ii) $2D_f(x, y) \leq L \|x - y\|^2$ for all $x, y \in \mathbb{R}^d$

(iii) $\langle \nabla f(x) - \nabla f(y), x - y \rangle \leq L \|x - y\|^2$ for all $x, y \in \mathbb{R}^d$

Let $g(x) = \frac{L}{2} \|x\|^2 - f(x)$, so $\nabla g(x) = Lx - \nabla f(x)$.

(i) $\Rightarrow$ (ii)

If f is L-smooth, then g is convex, which means that (Proposition 7)

$$2D_g(x, y) = L \|x\|^2 - L \|y\|^2 - 2f(x) + 2f(y) + 2 \langle \nabla f(y), x - y \rangle - 2L \langle y, x - y \rangle \geq 0,$$

which means

$$2f(x) - 2f(y) - 2 \langle \nabla f(y), x - y \rangle \leq L\big(\|x\|^2 - \|y\|^2 - 2 \langle y, x - y \rangle\big) = L \|x - y\|^2,$$

which is the same as $2D_f(x, y) \leq L \|x - y\|^2$.

(ii) ⇒ (iii)

By the computation above, $2D_g(x,y) = L\|x-y\|^2 - 2D_f(x,y)$. If $2D_f(x,y) \leq L\|x-y\|^2$, then $D_g(x,y) \geq 0$, then (adding $D_g(x,y) \geq 0$ and $D_g(y,x) \geq 0$)

$$\langle \nabla g(x) - \nabla g(y), x-y \rangle \geq 0,$$

which is the same as

$$L\langle x-y, x-y\rangle - \langle \nabla f(x) - \nabla f(y), x-y\rangle \geq 0,$$

and as a result $\langle \nabla f(x) - \nabla f(y), x-y\rangle \leq L\|x-y\|^2$.

(iii) ⇒ (i)

If $\langle \nabla f(x) - \nabla f(y), x-y\rangle \leq L\|x-y\|^2$, then $\langle \nabla g(x) - \nabla g(y), x-y\rangle \geq 0$, which means g is convex (Proposition 7, (iii) ⇒ (i)), which means that f is L-smooth.

Exercise 18

If $f : \mathbb{R}^d \to \mathbb{R}$, $f \in C^2(\mathbb{R}^d)$, then f is L-smooth iff $\nabla^2 f(x) \preceq LI$.

1)

If f is L-smooth, then $g(x) := \frac{L}{2}\|x\|^2 - f(x)$ is convex and $g \in C^2(\mathbb{R}^d)$, as a result (Exercise 12) $\nabla^2 g(x) = LI - \nabla^2 f(x) \succeq 0$, as a result $\nabla^2 f(x) \preceq LI$.

2)

If $\nabla^2 f(x) \preceq LI$, then $\nabla^2 g(x) \succeq 0$, as a result g is convex (Exercise 12), so f is L-smooth.